Interacting Multipoles of Rank 3 and 4

$$
\left[\begin{array}{c}
-135135 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot R_{\eta} \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta\left(\zeta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\zeta} \cdot \delta\left(\epsilon,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\eta} \cdot \delta\left(\epsilon,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot \delta\left(\delta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot R_{\eta} \cdot \delta\left(\delta,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\delta,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot \delta\left(\gamma,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\eta} \cdot \delta\left(\gamma,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\gamma,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\gamma,\delta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot \delta\left(\beta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\eta} \cdot \delta\left(\beta,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\beta,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\beta,\delta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\beta,\gamma\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot \delta\left(\alpha,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\eta} \cdot \delta\left(\alpha,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\alpha,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\alpha,\delta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\alpha,\gamma\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot R_{\zeta} \cdot R_{\eta} \cdot \delta\left(\alpha,\beta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta\left(\delta,\epsilon\right) \cdot \delta\left(\zeta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta\left(\delta,\zeta\right) \cdot \delta\left(\epsilon,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta\left(\delta,\eta\right) \cdot \delta\left(\epsilon,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta\left(\gamma,\epsilon\right) \cdot \delta\left(\zeta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta\left(\gamma,\zeta\right) \cdot \delta\left(\epsilon,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta\left(\gamma,\eta\right) \cdot \delta\left(\epsilon,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta\left(\gamma,\delta\right) \cdot \delta\left(\zeta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta\left(\gamma,\zeta\right) \cdot \delta\left(\delta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta\left(\gamma,\eta\right) \cdot \delta\left(\delta,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\zeta} \cdot \delta\left(\gamma,\delta\right) \cdot \delta\left(\epsilon,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\zeta} \cdot \delta\left(\gamma,\epsilon\right) \cdot \delta\left(\delta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\zeta} \cdot \delta\left(\gamma,\eta\right) \cdot \delta\left(\delta,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\eta} \cdot \delta\left(\gamma,\delta\right) \cdot \delta\left(\epsilon,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\eta} \cdot \delta\left(\gamma,\epsilon\right) \cdot \delta\left(\delta,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\eta} \cdot \delta\left(\gamma,\zeta\right) \cdot \delta\left(\delta,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta\left(\beta,\epsilon\right) \cdot \delta\left(\zeta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta\left(\beta,\zeta\right) \cdot \delta\left(\epsilon,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta\left(\beta,\eta\right) \cdot \delta\left(\epsilon,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta\left(\beta,\delta\right) \cdot \delta\left(\zeta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta\left(\beta,\zeta\right) \cdot \delta\left(\delta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta\left(\beta,\eta\right) \cdot \delta\left(\delta,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\zeta} \cdot \delta\left(\beta,\delta\right) \cdot \delta\left(\epsilon,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\zeta} \cdot \delta\left(\beta,\epsilon\right) \cdot \delta\left(\delta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\zeta} \cdot \delta\left(\beta,\eta\right) \cdot \delta\left(\delta,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\eta} \cdot \delta\left(\beta,\delta\right) \cdot \delta\left(\epsilon,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\eta} \cdot \delta\left(\beta,\epsilon\right) \cdot \delta\left(\delta,\zeta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\eta} \cdot \delta\left(\beta,\zeta\right) \cdot \delta\left(\delta,\epsilon\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta\left(\beta,\gamma\right) \cdot \delta\left(\zeta,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta\left(\beta,\zeta\right) \cdot \delta\left(\gamma,\eta\right) \cdot \Omega^{B}_{\alpha\beta\gamma} \cdot \Phi^{A}_{\delta\epsilon\zeta\eta}
\end{array}\right.
$$

Simplify deltas:

$$
\left[\begin{array}{c}
-135135 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\zeta,\zeta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\eta \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\eta \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\gamma,\gamma\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\eta \cdot \delta\left(\gamma,\gamma\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\gamma,\gamma\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\gamma,\gamma\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\beta,\beta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\eta \cdot \delta\left(\beta,\beta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\beta,\beta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\beta,\beta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\beta,\beta\right) \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\alpha,\alpha\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\eta \cdot \delta\left(\alpha,\alpha\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\alpha,\alpha\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\alpha,\alpha\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\alpha,\alpha\right) \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \delta\left(\alpha,\alpha\right) \cdot \Omega^B_{\alpha\alpha\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta\left(\delta,\delta\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta\left(\delta,\delta\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta\left(\delta,\delta\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\zeta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\zeta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\zeta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\eta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\eta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\eta \cdot \delta\left(\gamma,\gamma\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\zeta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\zeta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\zeta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\eta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\eta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\eta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\gamma\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\gamma\delta\epsilon}
\end{array}\right.
$$

$$
\begin{bmatrix}
-135135 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\alpha\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\eta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\gamma\delta\epsilon} \\
-945\ R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\gamma\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\gamma\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\gamma\delta\zeta}
\end{bmatrix}
$$

$$
\left[
\begin{array}{l}
-135135 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\zeta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\zeta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot R_\zeta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\gamma \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\beta \cdot \Omega^B_{\alpha\alpha\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\gamma\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\gamma\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\gamma\delta\epsilon} \\
-945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\gamma\delta\epsilon}
\end{array}
\right.
$$

$$
\xRightarrow{\text{added up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix}
-135135 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\eta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\zeta\zeta} \\
+20790 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\epsilon\epsilon\zeta} \\
+31185 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\zeta} \\
+41580 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+41580 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+41580 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\gamma\delta\epsilon\zeta} \\
+10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega^B_{\alpha\alpha\gamma} \cdot \Phi^A_{\beta\delta\epsilon\zeta} \\
-2835 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\delta\delta\epsilon\epsilon} \\
-2835 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\epsilon\epsilon} \\
-8505 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\gamma\delta\delta\epsilon} \\
-2835 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\epsilon\epsilon} \\
-8505 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\delta\delta\epsilon} \\
-5670 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\gamma\delta\epsilon} \\
-11340 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\gamma\delta\epsilon} \\
-2835 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\epsilon\epsilon} \\
-8505 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\delta\delta\epsilon} \\
-5670 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\gamma\gamma\delta\epsilon} \\
-11340 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\gamma\delta\epsilon} \\
-5670 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\alpha\gamma} \cdot \Phi^A_{\beta\beta\delta\epsilon} \\
-11340 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\beta\delta\epsilon} \\
-7560 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\beta\gamma\delta\epsilon} \\
-3780 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\alpha\gamma\delta\epsilon} \\
+315 \cdot R_z^6 \cdot R_\alpha \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\gamma\gamma\delta\delta} \\
+1260 \cdot R_z^6 \cdot R_\alpha \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\beta\gamma\delta\delta} \\
+315 \cdot R_z^6 \cdot R_\beta \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\gamma\gamma\delta\delta} \\
+1260 \cdot R_z^6 \cdot R_\beta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\gamma\delta\delta} \\
+315 \cdot R_z^6 \cdot R_\gamma \cdot \Omega^B_{\alpha\alpha\gamma} \cdot \Phi^A_{\beta\beta\delta\delta} \\
+1260 \cdot R_z^6 \cdot R_\gamma \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\beta\delta\delta} \\
+2520 \cdot R_z^6 \cdot R_\delta \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{\beta\gamma\gamma\delta} \\
+1260 \cdot R_z^6 \cdot R_\delta \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{\alpha\gamma\gamma\delta} \\
+2520 \cdot R_z^6 \cdot R_\delta \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{\alpha\beta\gamma\delta}
\end{bmatrix} \tag{2d}
$$

Simplify R:

$$
+\frac{1}{1575}\cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega^B_{\alpha\beta\gamma}\Phi^A_{\delta\epsilon\zeta\eta} = R_z^{-15}\cdot\frac{1}{1575}\cdot\left[\begin{array}{c}
-135135\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zzz}\cdot\Phi^A_{zzzz}\\
+10395\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zzz}\cdot\Phi^A_{zz\zeta\zeta}\\
+20790\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zzz}\cdot\Phi^A_{zz\epsilon\epsilon}\\
+31185\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zzz}\cdot\Phi^A_{zz\delta\delta}\\
+41580\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\gamma}\cdot\Phi^A_{zzz\gamma}\\
+41580\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\beta}\cdot\Phi^A_{zzz\beta}\\
+10395\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\beta\beta}\cdot\Phi^A_{zzzz}\\
+41580\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\alpha}\cdot\Phi^A_{zzz\alpha}\\
+10395\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{zzzz}\\
+10395\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{zzzz}\\
-2835\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zzz}\cdot\Phi^A_{\delta\delta\epsilon\epsilon}\\
-2835\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\gamma}\cdot\Phi^A_{z\gamma\epsilon\epsilon}\\
-8505\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\gamma}\cdot\Phi^A_{z\gamma\delta\delta}\\
-2835\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\beta}\cdot\Phi^A_{z\beta\epsilon\epsilon}\\
-8505\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\beta}\cdot\Phi^A_{z\beta\delta\delta}\\
-5670\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\beta\beta}\cdot\Phi^A_{zz\gamma\gamma}\\
-11340\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\beta\gamma}\cdot\Phi^A_{zz\beta\gamma}\\
-2835\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\alpha}\cdot\Phi^A_{z\alpha\epsilon\epsilon}\\
-8505\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{zz\alpha}\cdot\Phi^A_{z\alpha\delta\delta}\\
-5670\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{zz\gamma\gamma}\\
-11340\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\alpha\gamma}\cdot\Phi^A_{zz\alpha\gamma}\\
-5670\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{zz\beta\beta}\\
-11340\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{z\alpha\beta}\cdot\Phi^A_{zz\alpha\beta}\\
-7560\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{\alpha\alpha\beta}\cdot\Phi^A_{zzz\beta}\\
-3780\cdot R_z^4\cdot R_z\cdot R_z\cdot R_z\cdot\Omega^B_{\alpha\beta\beta}\cdot\Phi^A_{zzz\alpha}\\
+315\cdot R_z^6\cdot R_z\cdot\Omega^B_{z\beta\beta}\cdot\Phi^A_{\gamma\gamma\delta\delta}\\
+1260\cdot R_z^6\cdot R_z\cdot\Omega^B_{z\beta\gamma}\cdot\Phi^A_{\beta\gamma\delta\delta}\\
+315\cdot R_z^6\cdot R_z\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{\gamma\gamma\delta\delta}\\
+1260\cdot R_z^6\cdot R_z\cdot\Omega^B_{z\alpha\gamma}\cdot\Phi^A_{\alpha\gamma\delta\delta}\\
+315\cdot R_z^6\cdot R_z\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{\beta\beta\delta\delta}\\
+1260\cdot R_z^6\cdot R_z\cdot\Omega^B_{z\alpha\beta}\cdot\Phi^A_{\alpha\beta\delta\delta}\\
+2520\cdot R_z^6\cdot R_z\cdot\Omega^B_{\alpha\alpha\beta}\cdot\Phi^A_{z\beta\gamma\gamma}\\
+1260\cdot R_z^6\cdot R_z\cdot\Omega^B_{\alpha\beta\beta}\cdot\Phi^A_{z\alpha\gamma\gamma}\\
+2520\cdot R_z^6\cdot R_z\cdot\Omega^B_{\alpha\beta\gamma}\cdot\Phi^A_{z\alpha\beta\gamma}
\end{array}\right] \tag{3a}
$$

$$
\xRightarrow{\text{cleaned up}} R_z^{-15}\cdot\frac{1}{1575}\cdot\left[\begin{array}{c}
+41580\cdot R_z^7\cdot\Omega^B_{zz\gamma}\cdot\Phi^A_{zzz\gamma}+41580\cdot R_z^7\cdot\Omega^B_{zz\beta}\cdot\Phi^A_{zzz\beta}\\
+41580\cdot R_z^7\cdot\Omega^B_{zz\alpha}\cdot\Phi^A_{zzz\alpha}-2835\cdot R_z^7\cdot\Omega^B_{zz\gamma}\cdot\Phi^A_{z\gamma\epsilon\epsilon}\\
-8505\cdot R_z^7\cdot\Omega^B_{zz\gamma}\cdot\Phi^A_{z\gamma\delta\delta}-2835\cdot R_z^7\cdot\Omega^B_{zz\beta}\cdot\Phi^A_{z\beta\epsilon\epsilon}\\
-8505\cdot R_z^7\cdot\Omega^B_{zz\beta}\cdot\Phi^A_{z\beta\delta\delta}-11340\cdot R_z^7\cdot\Omega^B_{z\beta\gamma}\cdot\Phi^A_{zz\beta\gamma}\\
-2835\cdot R_z^7\cdot\Omega^B_{zz\alpha}\cdot\Phi^A_{z\alpha\epsilon\epsilon}\\
-8505\cdot R_z^7\cdot\Omega^B_{zz\alpha}\cdot\Phi^A_{z\alpha\delta\delta}-11340\cdot R_z^7\cdot\Omega^B_{z\alpha\gamma}\cdot\Phi^A_{zz\alpha\gamma}\\
-11340\cdot R_z^7\cdot\Omega^B_{z\alpha\beta}\cdot\Phi^A_{zz\alpha\beta}-7560\cdot R_z^7\cdot\Omega^B_{\alpha\alpha\beta}\cdot\Phi^A_{zzz\beta}\\
-3780\cdot R_z^7\cdot\Omega^B_{\alpha\beta\beta}\cdot\Phi^A_{zzz\alpha}+1260\cdot R_z^7\cdot\Omega^B_{z\beta\gamma}\cdot\Phi^A_{\beta\gamma\delta\delta}\\
+1260\cdot R_z^7\cdot\Omega^B_{z\alpha\gamma}\cdot\Phi^A_{\alpha\gamma\delta\delta}\\
+1260\cdot R_z^7\cdot\Omega^B_{z\alpha\beta}\cdot\Phi^A_{\alpha\beta\delta\delta}\\
+2520\cdot R_z^7\cdot\Omega^B_{\alpha\alpha\beta}\cdot\Phi^A_{z\beta\gamma\gamma}\\
+1260\cdot R_z^7\cdot\Omega^B_{\alpha\beta\beta}\cdot\Phi^A_{z\alpha\gamma\gamma}\\
+2520\cdot R_z^7\cdot\Omega^B_{\alpha\beta\gamma}\cdot\Phi^A_{z\alpha\beta\gamma}
\end{array}\right] \tag{3b}
$$

$$\xRightarrow{\text{reduced indices}} R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} +41580 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{zzz\alpha} + 41580 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{zzz\alpha} \\ +41580 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{zzz\alpha} - 2835 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{z\alpha\beta\beta} \\ -8505 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{z\alpha\beta\beta} - 2835 \cdot R_z^7 \cdot \Omega^B_{zz\beta} \cdot \Phi^A_{z\alpha\alpha\beta} \\ -8505 \cdot R_z^7 \cdot \Omega^B_{zz\beta} \cdot \Phi^A_{z\alpha\alpha\beta} - 11340 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} \\ -2835 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{z\alpha\beta\beta} - 8505 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{z\alpha\beta\beta} \\ -11340 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} - 11340 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} \\ -7560 \cdot R_z^7 \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{zzz\beta} - 3780 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{zzz\alpha} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\beta\gamma} \cdot \Phi^A_{\alpha\alpha\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\gamma} \cdot \Phi^A_{\alpha\beta\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{\alpha\beta\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{z\beta\gamma\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{z\alpha\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{z\alpha\beta\gamma} \end{bmatrix} \tag{3c}$$

$$\xRightarrow{\text{added up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} +124740 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{zzz\alpha} - 22680 \cdot R_z^7 \cdot \Omega^B_{zz\alpha} \cdot \Phi^A_{z\alpha\beta\beta} \\ -11340 \cdot R_z^7 \cdot \Omega^B_{zz\beta} \cdot \Phi^A_{z\alpha\alpha\beta} - 34020 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} \\ -7560 \cdot R_z^7 \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{zzz\beta} - 3780 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{zzz\alpha} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\beta\gamma} \cdot \Phi^A_{\alpha\alpha\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\gamma} \cdot \Phi^A_{\alpha\beta\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{\alpha\beta\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\alpha\beta} \cdot \Phi^A_{z\beta\gamma\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\beta} \cdot \Phi^A_{z\alpha\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{z\alpha\beta\gamma} \end{bmatrix} \tag{3d}$$

Simplify Oktopoles:

$$+\frac{1}{1575} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega^B_{\alpha\beta\gamma}\Phi^A_{\delta\epsilon\zeta\eta} = R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} +124740 \cdot R_z^7 \cdot \Omega^B_{yzz} \cdot \Phi^A_{yzzz} - 22680 \cdot R_z^7 \cdot \Omega^B_{yzz} \cdot \Phi^A_{yz\beta\beta} \\ -11340 \cdot R_z^7 \cdot \Omega^B_{yzz} \cdot \Phi^A_{yz\alpha\alpha} - 34020 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} \\ -7560 \cdot R_z^7 \cdot \Omega^B_{y\alpha\alpha} \cdot \Phi^A_{yzzz} - 3780 \cdot R_z^7 \cdot \Omega^B_{y\beta\beta} \cdot \Phi^A_{yzzz} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\beta\gamma} \cdot \Phi^A_{\alpha\alpha\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\gamma} \cdot \Phi^A_{\alpha\beta\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{\alpha\beta\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{y\alpha\alpha} \cdot \Phi^A_{yz\gamma\gamma} + 1260 \cdot R_z^7 \cdot \Omega^B_{y\beta\beta} \cdot \Phi^A_{yz\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{z\alpha\beta\gamma} \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} -34020 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} + 1260 \cdot R_z^7 \cdot \Omega^B_{z\beta\gamma} \cdot \Phi^A_{\alpha\alpha\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\gamma} \cdot \Phi^A_{\alpha\beta\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{\alpha\beta\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{z\alpha\beta\gamma} \end{bmatrix} \tag{4b}$$

$$\xRightarrow{\text{reduced indices}} R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} -34020 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} + 1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{\alpha\beta\gamma\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\gamma} \cdot \Phi^A_{\alpha\beta\beta\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{\alpha\beta\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{z\alpha\beta\gamma} \end{bmatrix} \tag{4c}$$

$$\xRightarrow{\text{added up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} -34020 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{zz\alpha\beta} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{z\alpha\beta} \cdot \Phi^A_{\alpha\beta\gamma\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega^B_{z\alpha\gamma} \cdot \Phi^A_{\alpha\beta\beta\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega^B_{\alpha\beta\gamma} \cdot \Phi^A_{z\alpha\beta\gamma} \end{bmatrix} \tag{4d}$$

Simplify Hexadecapoles:

$$+\frac{1}{1575}\cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega^B_{\alpha\beta\gamma}\Phi^A_{\delta\epsilon\zeta\eta} = R_z^{-15}\cdot\frac{1}{1575}\cdot\begin{bmatrix} -34020\cdot R_z^7\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{zz\alpha\alpha} \\ +2520\cdot R_z^7\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{\alpha\alpha\gamma\gamma} \\ +1260\cdot R_z^7\cdot\Omega^B_{z\alpha\alpha}\cdot\Phi^A_{\alpha\alpha\beta\beta} \\ +2520\cdot R_z^7\cdot\Omega^B_{\alpha\beta\gamma}\cdot\Phi^A_{z\alpha\beta\gamma}\end{bmatrix} \tag{5a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-15}\cdot\frac{1}{1575}\cdot\left[\ +2520\cdot R_z^7\cdot\Omega^B_{\alpha\beta\gamma}\cdot\Phi^A_{z\alpha\beta\gamma}\ \right] \tag{5b}$$

Multiply Out Sum:

$$+\frac{1}{1575}\cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega^B_{\alpha\beta\gamma}\Phi^A_{\delta\epsilon\zeta\eta} = R_z^{-15}\cdot\frac{1}{1575}\cdot\begin{bmatrix} +2520\cdot R_z^7\cdot\Omega^B_{xxx}\cdot\Phi^A_{xxxz} + 2520\cdot R_z^7\cdot\Omega^B_{xxy}\cdot\Phi^A_{xxyz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xxz}\cdot\Phi^A_{xxzz} + 2520\cdot R_z^7\cdot\Omega^B_{xxy}\cdot\Phi^A_{xxyz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xyy}\cdot\Phi^A_{xyyz} + 2520\cdot R_z^7\cdot\Omega^B_{xyz}\cdot\Phi^A_{xyzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xxz}\cdot\Phi^A_{xxzz} + 2520\cdot R_z^7\cdot\Omega^B_{xyz}\cdot\Phi^A_{xyzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xzz}\cdot\Phi^A_{xzzz} + 2520\cdot R_z^7\cdot\Omega^B_{xxy}\cdot\Phi^A_{xxyz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xyy}\cdot\Phi^A_{xyyz} + 2520\cdot R_z^7\cdot\Omega^B_{xyz}\cdot\Phi^A_{xyzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xyy}\cdot\Phi^A_{xyyz} + 2520\cdot R_z^7\cdot\Omega^B_{yyy}\cdot\Phi^A_{yyyz} \\ +2520\cdot R_z^7\cdot\Omega^B_{yyz}\cdot\Phi^A_{yyzz} + 2520\cdot R_z^7\cdot\Omega^B_{xyz}\cdot\Phi^A_{xyzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{yyz}\cdot\Phi^A_{yyzz} + 2520\cdot R_z^7\cdot\Omega^B_{yzz}\cdot\Phi^A_{yzzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xxz}\cdot\Phi^A_{xxzz} + 2520\cdot R_z^7\cdot\Omega^B_{xyz}\cdot\Phi^A_{xyzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xzz}\cdot\Phi^A_{xzzz} + 2520\cdot R_z^7\cdot\Omega^B_{xyz}\cdot\Phi^A_{xyzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{yyz}\cdot\Phi^A_{yyzz} + 2520\cdot R_z^7\cdot\Omega^B_{yzz}\cdot\Phi^A_{yzzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{xzz}\cdot\Phi^A_{xzzz} + 2520\cdot R_z^7\cdot\Omega^B_{yzz}\cdot\Phi^A_{yzzz} \\ +2520\cdot R_z^7\cdot\Omega^B_{zzz}\cdot\Phi^A_{zzzz}\end{bmatrix} \tag{6a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-15}\cdot\frac{1}{1575}\cdot[+0] \tag{6b}$$

Combining everything:

$$+\frac{1}{1575}\cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega^B_{\alpha\beta\gamma}\Phi^A_{\delta\epsilon\zeta\eta} = +0 \tag{7}$$